

GOODRIDGE 12 NNW, MN, USA

WMO: 726470

Lat: 48.3055N

Lon: 95.8744W

Elev: 1150

StdP: 14.10

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 04994

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.7	(c) -20.2	(d) -30.5	(e) 1.0	(f) -24.6	(g) -25.9	(h) 1.4	(i) -19.9	(j) 15.8	(k) 7.2	(l) 13.9	(m) 10.7	(n) 2.0	(o) N/A	(p) 0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	84.8	69.7	82.0	67.5	79.5	66.0	72.6	81.7	70.3	79.1	68.3	76.8	5.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.2	112.0	78.7	67.1	103.9	76.4	65.0	96.7	73.9	36.9	82.1	34.9	79.4	33.2	76.7	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	11.0	9.7	-33.6	90.2	6.3	3.1	-38.1	92.5	-41.8	94.3	-45.4	96.1	-50.0	98.3
			-31.9	76.4	6.9	1.9	-36.8	77.8	-40.9	78.9	-44.8	80.0	-49.8	81.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	5.5	7.8	24.0	39.7	52.2	61.8	66.1	63.1	55.7	42.7	26.6	11.8
	DBStd	23.89	15.56	13.50	13.63	10.32	8.95	6.11	5.58	6.25	8.53	9.07	11.73	13.10
	HDD50	5976	1380	1182	808	331	81	2	0	1	38	266	702	1185
	HDD65	9971	1845	1602	1270	760	407	130	54	113	296	693	1151	1650
	CDD50	1684	0	0	4	21	149	355	499	406	209	40	1	0
	CDD65	204	0	0	0	0	10	33	88	53	18	2	0	0
	CDH74	2292	0	0	0	5	155	379	935	578	213	27	0	0
	CDH80	466	0	0	0	0	34	58	208	120	42	4	0	0
Wind	WSAvg	4.2	4.5	4.4	4.6	4.9	4.7	3.9	3.3	3.2	3.8	4.2	4.4	4.3
Precipitation	PrecAvg	23.6	0.6	0.5	0.8	1.4	3.1	3.9	3.8	3.5	2.5	1.9	1.0	0.7
	PrecMax	28.4	1.1	0.9	2.6	2.6	7.2	8.9	6.1	5.7	5.4	5.1	3.3	1.4
	PrecMin	16.6	0.2	0.1	0.2	0.6	0.9	0.9	0.9	0.9	0.5	0.3	0.1	0.3
	PrecStd	3.7	0.3	0.3	0.6	0.6	1.7	2.0	1.4	1.3	1.4	1.2	0.9	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	41.3	64.6	72.9	84.4	85.3	88.4	87.5	84.8	78.1	58.9	39.8
		MCWB	35.0	37.0	54.2	51.4	63.6	68.6	72.7	71.5	67.8	59.4	48.4	36.9
	2%	DB	34.2	35.4	54.1	67.2	78.4	81.3	85.1	83.4	80.0	69.2	51.7	35.4
		MCWB	32.1	32.6	46.4	51.0	59.8	65.5	71.9	68.6	64.6	55.2	44.5	32.3
	5%	DB	31.2	32.3	46.4	62.3	73.5	78.4	82.1	80.0	75.7	62.7	46.8	32.5
		MCWB	29.4	30.1	40.4	48.1	56.8	63.9	69.6	67.5	62.8	52.4	41.7	30.6
	10%	DB	27.6	28.2	41.1	57.8	69.1	75.5	79.4	77.0	71.1	57.7	42.0	29.5
		MCWB	25.8	26.2	36.3	45.8	54.8	62.6	67.6	65.5	60.8	49.2	38.2	27.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.4	38.3	58.7	54.2	66.3	72.0	76.4	74.1	71.5	60.2	49.9	38.3
		MCDB	39.0	42.9	64.1	64.2	79.2	79.4	84.2	85.2	79.1	75.7	57.3	39.8
	2%	WB	33.0	33.3	49.7	50.4	63.5	68.3	73.3	72.0	67.7	57.2	45.7	33.8
		MCDB	34.7	35.8	55.5	62.8	74.8	77.7	82.0	81.4	75.1	67.1	51.3	35.9
	5%	WB	29.9	30.5	42.1	47.9	60.3	66.1	71.0	69.5	64.6	53.7	41.5	31.3
		MCDB	32.0	32.7	48.4	59.6	69.6	76.0	79.7	77.9	72.6	61.2	46.0	33.4
	10%	WB	26.0	26.8	37.4	45.4	57.7	64.1	68.9	67.6	61.6	50.2	37.9	28.3
		MCDB	27.6	28.8	42.6	55.3	66.4	73.8	77.6	75.4	69.5	57.8	41.6	30.0
Mean Daily Temperature Range	5% DB	MDBR	18.4	19.7	19.1	21.2	22.3	20.9	22.5	23.5	22.4	18.4	15.8	15.5
		MCDBR	16.4	18.0	23.7	29.6	29.5	25.9	25.0	26.9	27.7	27.1	21.3	16.4
		MCWBR	15.6	15.3	16.5	16.9	14.9	14.0	13.5	14.5	15.5	14.7	14.2	12.9
	5% WB	MCDBR	17.5	17.0	24.0	28.0	23.4	23.3	21.9	23.5	23.4	22.7	19.7	14.5
		MCWBR	15.4	14.3	16.7	16.7	13.3	13.6	13.3	13.5	14.3	13.3	13.5	12.4
Clear-Sky Solar Irradiance	taub		0.271	0.306	0.294	0.346	0.362	0.376	0.379	0.377	0.348	0.303	0.276	0.263
	taud		2.281	2.175	2.415	2.366	2.381	2.367	2.372	2.383	2.459	2.553	2.507	2.378
	Ebn at Noon		253	266	295	287	285	281	278	273	272	271	256	244
	Edh at Noon		23	33	30	36	36	37	37	35	29	23	19	19
All-Sky Solar Radiation	RadAvg		473	858	1230	1531	1718	1834	1966	1640	1209	729	423	335
	RadStd		51	69	123	135	106	140	87	115	90	107	49	37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page